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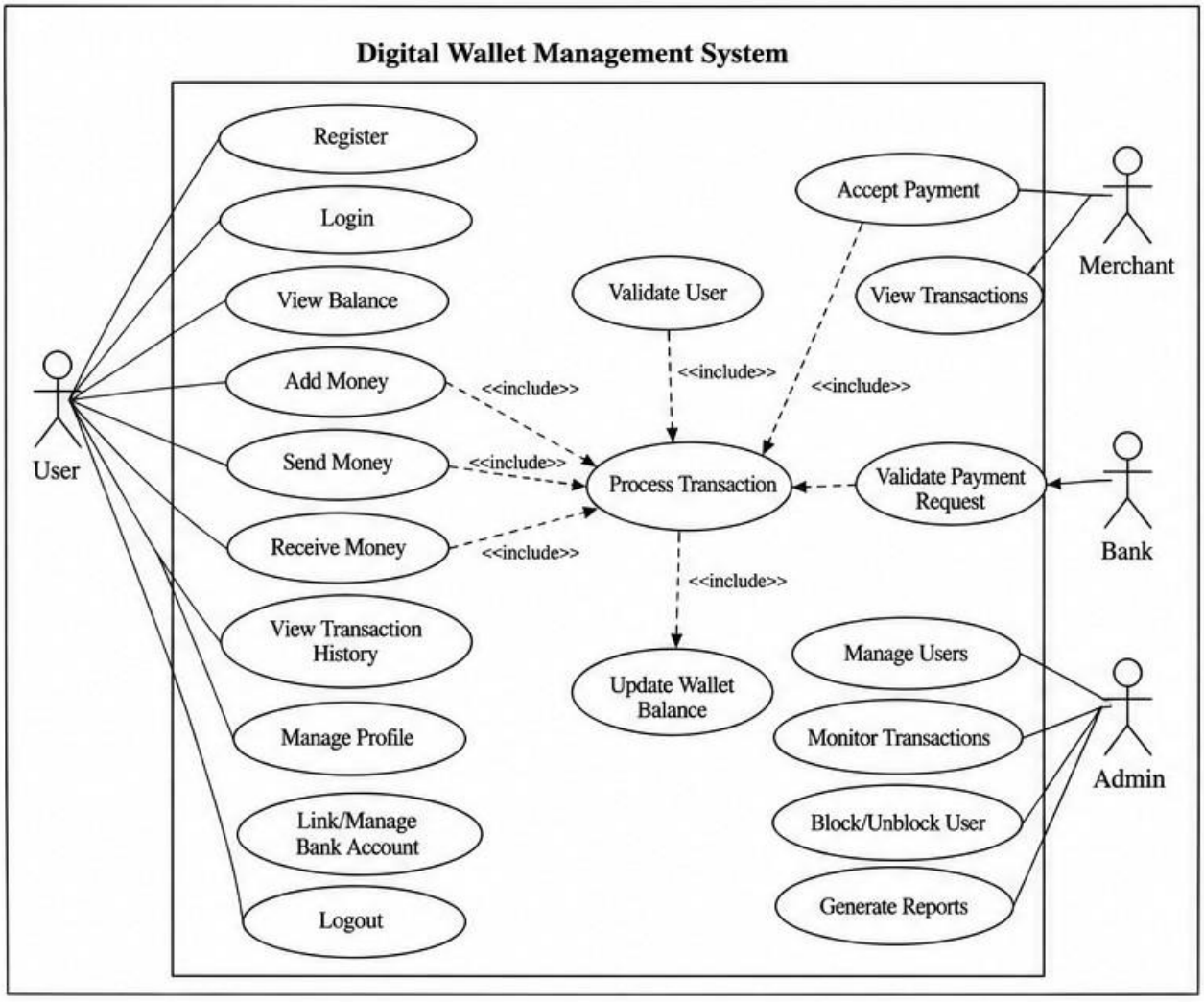
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Section : B

Subject : Software Engineering

# Digital Wallet Management System – Use Case Diagram



# Actors and Their Roles

| Actor | Role / Description |
|---|---|
| User | A registered customer who uses the digital wallet to manage money. The user can register, log in, add money, send and receive money, view balance and transaction history, manage the profile, and link a bank account. |
| Merchant | A business or vendor who accepts payments from users through the digital wallet, UPI or QR-based payment and can view transactions related to the merchant account. |
| Bank | An external banking system that validates payment requests, authorizes transactions, and supports fund transfers between the wallet and bank account. |
| Admin | The system administrator who manages users, monitors transactions, blocks or unblocks users, and generates system reports. |

# Use Case Descriptions

| Use Case | Actor(s) | Precondition | Postcondition | Function / Description |
|---|---|---|---|---|
| Register | User | User is not registered. | User account is created. | Creates a new user account with required personal details. |
| Login | User | User is registered and active. | User is authenticated and a session is started. | Authenticates the user and provides access to the system. |
| View Balance | User | User is logged in. | Current wallet balance is displayed. | Displays the current available wallet balance. |
| Add Money | User | User is logged in and has a linked payment method. | Wallet balance is increased. | Allows the user to add money using a bank account, card or UPI. |
| Send Money | User | User is logged in, has sufficient balance and valid receiver details. | Amount is transferred and wallet balance is updated. | Transfers money to another user or merchant. |
| Receive Money | User | Sender initiates a transfer and receiver is valid. | Amount is credited to receiver's wallet. | Allows the user to receive money from another user or merchant. |
| View Transaction History | User, Merchant | User/Merchant is logged in. | Transaction list is displayed. | Displays previous transactions with date, amount, type and status. |
| Manage Profile | User | User is logged in. | Profile information is updated. | Allows the user to view and update personal details. |
| Link/Manage Bank Account | User | User is logged in. | Bank account/payment method is linked or updated. | Allows the user to link, update or remove a bank account/payment method. |
| Logout | User | User is logged in. | User session is ended. | Logs the user out of the digital wallet system. |
| Accept Payment | Merchant | Merchant is logged in and has a valid merchant account. | Payment is received in merchant account. | Allows a merchant to accept payments through wallet, UPI or QR. |
| View Transactions | Merchant | Merchant is logged in. | Merchant transaction list is displayed. | Shows payments received by the merchant. |
| Validate Payment Request | Bank | A payment request has been initiated. | Payment is validated and confirmation is sent. | Verifies the bank account/payment request and authorizes the transaction. |
| Process Transaction | System | User is authenticated and request is valid. | Transaction is processed and status is updated. | Handles validation, fund transfer and transaction processing. |
| Update Wallet Balance | System | Transaction is successfully processed. | Wallet balance is updated. | Updates the wallet balance after a successful transaction. |
| Manage Users | Admin | Admin is logged in. | User details are updated. | Allows the administrator to view, edit or manage user accounts. |
| Monitor Transactions | Admin | Admin is logged in. | Transaction logs are displayed. | Allows the administrator to monitor system transactions. |
| Block/Unblock User | Admin | Admin is logged in and user exists. | User is blocked or unblocked. | Allows the administrator to control user access. |
| Generate Reports | Admin | Admin is logged in. | System report is generated. | Generates transaction and system reports for administration. |